Research evolution

Rohan Alwyn | FIN439 Project 1 | October 19, 2026

Baseline provenance and the instructor's October 19 clarification

The recovered local artifact is titled "Project 1 Valuation - Edition A" and names September 3, 2026 as its valuation date. It is preserved byte-for-byte in prior-labs/evidence/Project_1_Valuation_Edition_A_NVDA.md. SHA-256: 05bb32f69ca7d0d2bed65a1c7273147662bfd1bb1f4e506aacae838dbc0b9462

The separate Brightspace Edition A checkpoint showed Not Submitted on October 19. The Lab 3 quiz is a separate object and shows 25/25. Neither that grade nor this local filename establishes a successful checkpoint receipt or AI-off authorship. This package does not claim either. Dr. Z's October 19 email directs students to preserve an available original unchanged, or honestly label a retrospective recollection if the original is unavailable. The local artifact below is presented with its actual provenance and uncertainty; it is not a reconstructed checkpoint. The comparison is to that recovered artifact, not a certified pre-AI baseline.

Edition B - decision and method

The user is the analyst group advising a fund with no NVIDIA position. The decision is initiate, watch/defer, or do not initiate over a five-year operating forecast. The proposed action remains watch/defer. September 24, 2026 is the frozen valuation date, with a $100-$285 rounded DCF scenario span and a $158 base against a $224.58 same-date close. October 19 is the preparation date, not the quote date.

Item	Recovered artifact / labs	Edition B	Reason and evidence
Decision	Watch/defer pending valuation work	Watch/defer, with price/value and risk gates	DCF scenarios and sensitivity support conditional restraint; unchanged recommendation is not unchanged evidence.
Cash-flow architecture	Edition A planned FCFF; Lab 10 implemented FCFE	Operating FCFF, WACC, full enterprise-equity bridge	Inspect model.py and primary debt/cash definitions. Cash-flow/rate matching is explicit.
Timing	Annual January anchor with September information	September 24 anchor, July reported BS, estimated roll and remaining stub	Elapsed cash flows removed; interim estimates labeled instead of treated as actuals.
Data	Earlier annual debt and basic shares	Q2 debt $33.366bn; diluted proxy 24.285bn; basic proxy separate	Q2 release balance sheet and EPS tables; old 24.1bn count was basic.
Quote	Saved $224.20 intraday reference	$224.58 September 24 close	Rechecked FinancialContent history; intraday and close are not interchangeable.
Peer check	September 3 P/E only	Same September 24 price date, P/E plus EV/sales	Recomputed from traced inputs; share/financial period staleness explicitly qualifies the references.
Validation	Accounting tests and old sensitivity output	Stub/bridge checks, corruption guard, directional tests and clean-venv run	Executed logs; numerical agreement within 1e-6 tolerance. Student locked record separately identified.

Consequential output check

The prior annual teaching value could not be interpreted as a September market comparison. The Q2 balance sheet, period-end dates and cash-flow identities support changing the timing architecture and bridge. Independent source evidence supports the correction; another AI's agreement does not. The model's technical checks demonstrate arithmetic consistency, not proof that forecasts or risk premia are correct.

What changed, and why?

The recommendation did not materially change, but its evidence did. The updated work now exposes the company's operating drivers, the difference between historical facts and forecasts, the cash-flow/rate pairing, the valuation-date estimates, market-implied discount rate and peer-definition conflicts. Course instruction, the prior lab work, source checks and AI-assisted implementation jointly drove the changes. There is no invented human partner exchange or claim that the complete comparison was performed before AI assistance. The student must review and own the final judgment and recordings.

Recovered local artifact - reproduced content

The following content is carried forward unchanged from the preserved Markdown source. The notice above governs its provenance; this is not proof of a checkpoint receipt.

Project 1 Valuation - Edition A

Decision and intended user

The intended user is an equity-research analyst deciding whether NVIDIA Corporation should advance to deeper valuation research and potential coverage initiation. The decision is whether to initiate detailed valuation work now, watch and defer pending additional evidence, or decline to initiate.

Company and valuation date

- Company: NVIDIA Corporation

- Ticker: NVDA

- Valuation date: September 3, 2026

- Initial thesis: **Watch and defer pending valuation work.** NVIDIA's growth, cash generation, and net-cash position justify deeper research, but this evidence alone does not establish that the shares are attractively valued.

Primary filing locator

- Filing: NVIDIA Corporation Form 10-K

- SEC accession: 0001045810-26-000021

- Reporting period: Fiscal year ended January 25, 2026

- Filing date: February 25, 2026

- SEC filing: https://www.sec.gov/Archives/edgar/data/1045810/000104581026000021/nvda-20260125.htm

- SEC filing-detail page:
https://www.sec.gov/Archives/edgar/data/1045810/000104581026000021/0001045810-26-000021-index.htm

- Currency: U.S. dollars

- Units: Millions, except per-share data where indicated

Known evidence

1. Fiscal 2026 revenue was $215.938 billion, up 65% from fiscal 2025. Data Center revenue increased 68%, making demand for accelerated computing and AI a central operating-value driver. Source: NVIDIA FY2026 Form 10-K, fiscal-year summary.

2. Net cash provided by operating activities was $102.718 billion in fiscal 2026. Capital expenditures were $6.1 billion. Source: NVIDIA FY2026 Form 10-K, Liquidity and Capital Resources.

3. Cash and cash equivalents were $10.605 billion, while cash, cash equivalents, and marketable securities totaled $62.556 billion as of January 25, 2026. Source: NVIDIA FY2026 Form 10-K, Liquidity and Capital Resources.

4. Debt had a net carrying amount of $8.468 billion, comprising $999 million short-term and $7.469 billion long-term. No commercial paper was outstanding as of January 25, 2026. Source: NVIDIA FY2026 Form 10-K, Outstanding Indebtedness and Commercial Paper Program.

5. Fiscal 2026 diluted weighted-average shares were 24.514 billion. This is an income-statement denominator incorporating the dilutive effect of equity awards under the treasury-stock method; it is not the period-end basic share count. Source: NVIDIA FY2026 Form 10-K, Note 4 - Net Income Per Share.

6. The Week 1 screen calculated a three-year revenue CAGR of 100.05% and a latest-year free-cash-flow margin of 44.77%. NVIDIA ranked first among the original qualifiers. Raising the revenue-CAGR threshold from 5% to 15% removed JNJ and GOOGL, but NVIDIA remained first ahead of META and MSFT. Source: Week 1 yfinance-based screening workbook, accessed September 3, 2026. These values require reconciliation to the primary filing before use in valuation.

7. Management concluded that disclosure controls and internal control over financial reporting were effective as of January 25, 2026. PricewaterhouseCoopers also gave an unqualified opinion on the financial statements and internal control. Source: NVIDIA FY2026 Form 10-K, Item 9A and auditor's report.

Consequential assumptions

- Recent AI-infrastructure demand will normalize from its exceptional growth rate but remain structurally positive.

- Revenue growth will slow materially from fiscal 2026 rather than continue at the historical screen's three-year CAGR.

- Gross margin will remain strong despite product transitions, export restrictions, inventory charges, and increasing competition.

- Capital expenditures and ecosystem investments will rise, so historical free-cash-flow conversion should not be carried forward unchanged.

- Cash and marketable securities are treated as non-operating assets in the enterprise-to-equity bridge, subject to confirming which securities are required for operations.

- Debt is measured using the filing's carrying amount initially; a later valuation should test whether market value differs materially.

- Diluted shares must reflect equity awards and future stock-based compensation rather than relying only on the basic period-end count.

Important unknowns

- The sustainable long-run growth rate for Data Center revenue after the current AI investment cycle.

- The effect of export controls, customer concentration, competition, and supply constraints on revenue and margins.

- A normalized operating margin and reinvestment rate across a full industry cycle.

- The appropriate weighted-average cost of capital and terminal-growth assumptions.

- The amount of stock-based compensation and future dilution that should be incorporated into valuation.

- Whether current market expectations already price in more growth than a defensible cash-flow forecast supports.

Research plan

1. Reconcile revenue, operating cash flow, capital expenditures, cash, debt, and diluted shares to the latest 10-K and subsequent 10-Q.

2. Build historical income-statement, balance-sheet, and cash-flow series using consistent fiscal periods and units.

3. Separate operating from non-operating assets and complete the enterprise-to-equity bridge.

4. Model revenue by major platform, with particular attention to Data Center demand, customer concentration, export restrictions, and supply.

5. Forecast operating margins, taxes, reinvestment, and FCFF under base, upside, and downside cases.

6. Estimate WACC and terminal value, then test sensitivity to growth, margins, reinvestment, and discount rate.

7. Compare implied value with the market price as of the valuation date before making any coverage-initiation conclusion.

Synthetic enterprise-to-equity bridge

The supplied mini-case uses USD millions except for the per-share result:

- NOPAT = $150 x (1 - 0.25) = **$112.50 million**

- FCFF = $112.50 + $40 - $60 - $15 = **$77.50 million**

- Common equity value = $1,000 + $100 - $250 - $20 - $10 = **$820.00 million**

- Value per diluted share = $820 / 100 = **$8.20 per share**

This is a synthetic known-answer check and is not an NVIDIA valuation.

Falsification question

Partner-generated question: Can hyperscalers establish a profitable, cash-generative ROI on their AI infrastructure investments before custom ASICs and merchant alternatives permanently compress NVIDIA's Data Center gross margins?

Why it matters: The thesis depends on customers earning enough from AI infrastructure to sustain their capital spending while NVIDIA preserves pricing power. If hyperscalers cannot generate adequate returns, their demand may slow; if custom ASICs or merchant alternatives become economically superior first, NVIDIA could lose share or face lasting margin compression. Either outcome would undermine the growth and cash-flow assumptions supporting deeper valuation research.